



Structural supercapacitors that double as protective shells

Engineers at the University of California San Diego have designed a structural supercapacitor. Researchers developed a solar-powered boat using a structural supercapacitor as its hull, with the motor powered by sunlight captured by a connected solar cell.



This miniature boat is built from a structural supercapacitor—a material providing structural support and energy storage capabilities (Credit: UC San Diego Jacobs School of Engineering)

This supercapacitor combines standard components but stands out due to its enhanced materials, including woven carbon fibre electrodes coated with a conductive polymer

and graphene oxide blend. A unique design feature is the varied concentration of polyethylene oxide in the electrolyte, denser near electrodes, ensuring optimal ion flow and maintaining structural integrity. This gradient design boosts electrolyte performance, offering better electrical properties on the edges and mechanical robustness in the centre. Supercapacitors provide rapid energy bursts but have less energy density than batteries; the team aims to improve this density and optimise both energy and power levels.

Efficient solar cell revolutionising green hydrogen

Researchers at the University of Tübingen have developed a highly efficient solar cell capable of producing hydrogen on a decentralised scale, with potential for large-scale deployment. This solar cell directly integrates with catalysts for water splitting in a photoelectrochemical setup, eliminating the need for an external circuit and enhancing cost-effectiveness. Despite potential efficiency and stability challenges due to crystalline defects that may arise during layer formation, this design achieved an 18% efficiency in direct solar water splitting, closely following global records. This technology enables solar hydrogen production with reduced reliance on the grid, advocating for decentralised energy solutions.

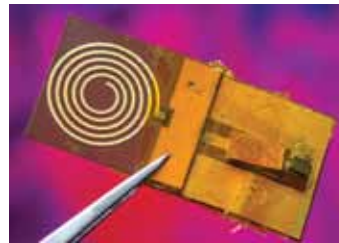
AI enhances safety for autonomous drone aircraft traffic

Researchers at the Institute for Assured Autonomy, led by Lanier Watkins and Louis Whitcomb, have harnessed artificial intelligence (AI) to develop a system for autonomously managing drone traffic, reducing human intervention. To make the

simulation more realistic, the team introduced ‘noisy sensors’ to mimic real-world unpredictability and employed a ‘fuzzy interference system’ to assess drone risk factors based on various parameters, enabling the system to autonomously prevent collisions. The research included various scenarios involving ‘rogue drones,’ and yielded promising results.

Revolutionising force detection with electronic stickers

University of California San Diego engineers have crafted electronic ‘stickers’ capable of gauging the force between two objects. These stickers are wireless, battery-free, and designed for confined spaces, making them adaptable for



This so-called “force sticker” is a thin, flexible electronic device that measures forces between objects in contact (Credit: David Baillot/UC San Diego Jacobs School of Engineering)

various uses. The force stickers, composed of a small capacitor and an RFID sticker, function like wireless barcodes using radio signals to transmit force data to an RFID reader. The stickers can detect compressive forces by measuring changes in their capacitance, with the sticker’s RFID component

reflecting the data. When applied to a variety of items, these stickers enabled users to gauge pressure, including squeezing, tapping, and pinching. Researchers envision applications in packaging, quality control, and robotics, where tight spaces or limited power sources restrict the use of conventional sensors.

AI for business operations and planning

Fujitsu has introduced a groundbreaking technology for automated AI solution generation, offering users the ability to customise AI innovation components. This technology is delivered through the ‘Fujitsu Kozuchi—Fujitsu AI Platform.’ The platform interprets programs and mathematical expressions from large language models (LLMs), producing graphical solutions that facilitate the creation of expert mathematical expressions. When AI models are trained using this graph data, they can automatically generate various AI models for tasks such as prediction, optimisation, and anomaly detection. By incorporating prior learning data into the graph, AI models can be efficiently adapted to new conditions. When combined with an LLM, this technology empowers users to quickly prototype, modify, and tailor AI solutions using natural language commands.